

Assignment 1

Part I

Q1. A traditional file system stores data without any particular structure or access control. A Database Management System (DBMS) provides structured data storage, a query language, transaction management and access control.

Consider a hospital managing patient records across multiple departments. With a file system, each department might maintain its own files, leading to data duplication, inconsistency, and no protection against two staff members overwriting the same record simultaneously. There might also be personal patient information which should be protected. A DBMS handles access via locking and transactions, ensures every record satisfies constraints (e.g. a patient ID must exist before a prescription is added), and allows complex queries across departments without duplicating data. This makes a DBMS clearly the better choice.

Q2. A strong entity exists independently. A weak entity does not have a primary key of its own and depends on a related strong entity for its existence and identification.

The primary key of a weak entity is formed by combining the primary key of its owner entity with the weak entity's own partial key (discriminator).

The relationship linking a weak entity to its owner is called an identifying relationship. It is typically represented in ER diagrams with a double-lined diamond.

Q3. Multivalued attribute

A multivalued attribute is one that can hold multiple values for a single entity instance, which is the case here as one student may have zero, one, or several phone numbers

Q4. Super key: Any set of one or more attributes that uniquely identifies a row in a relation. A super key may contain extra (redundant) attributes.

Candidate key: A super key from which no attribute can be removed while still preserving uniqueness. Every candidate key is a super key, but not every super key is a candidate key.

For the relation Student(roll_no, email, name, branch):

Candidate keys:

- roll_no - uniquely identifies each student; no subset (only itself) still identifies uniquely.
- email - each student has a unique email address; minimal.

Super keys that are not candidate keys:

- {roll_no, email} — unique, but removing either attribute still gives a unique identifier, so it is not minimal.
- {roll_no, name} — unique (roll_no alone suffices), so name is redundant; not minimal.

Q5. Constraint violated: Foreign key constraint

What it protects against: It ensures that every value in a foreign key column corresponds to an existing primary key value in the parent table.

What should happen: The insert operation should be rejected (rolled back) by the DBMS. The database returns a constraint violation error and the new row is not added to the Enrollment table.

Q6. A junction table must be created to represent the M:N relationship.

The primary key of the junction table consists of the combination of the primary keys of both participating entity tables (a composite primary key).

In a relational model, a column holds exactly one value per row. If we tried to store a many-to-many link as a column in one entity's table, a single row would need to reference multiple rows in the other table which is not possible.

Part II

Entities and Key Attributes

- Customer: customer_id , name, email, phone, delivery_address
- Restaurant: restaurant_id , name, address, contact_number
- MenuItem: item_id , name, description, price
- Order: order_id, order_date, total_amount, status

A) Customer → Order (1:N): A customer can place many orders, but each order belongs to exactly one customer.

Restaurant → Order (1:N): A restaurant can receive many orders, but each order is placed with exactly one restaurant.

Restaurant → MenuItem (1:N): A restaurant offers many menu items, but each menu item belongs to one restaurant.

Order – MenuItem (M:N): An order can contain multiple menu items, and the same menu item can appear in many different orders.

B) Multivalued attribute: phone on Customer. A customer may have more than one phone number.

Composite attribute: delivery_address on Customer. An address is naturally composed of distinct sub-parts: street_number, street_name, city, state, and pin_code. Because it can be meaningfully decomposed into component attributes, it qualifies as a composite attribute.

C) The M:N relationship between Order and MenuItem requires a junction table. Each row records one menu item within one order, along with the quantity ordered.

OrderItem(order_id, item_id, quantity, price)

Primary key: (order_id, item_id) - the combination uniquely identifies each line item within an order.

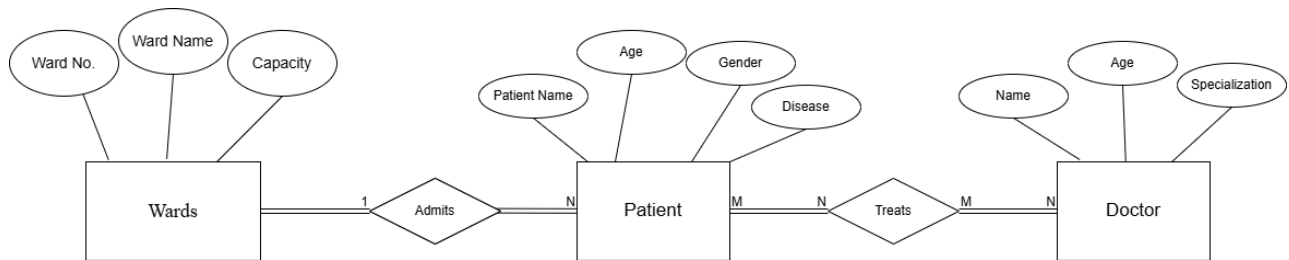
Foreign keys: order_id → Order(order_id); item_id → MenuItem(item_id)

Additional attribute: quantity (INTEGER, NOT NULL) - records how many units of the menu item are included in that order, as required by the scenario.

Price- price of the menu item

Part III

Q1.



Q2. Doctor–Patient M:N relationship: The relationship is M:N because a single doctor manages multiple patients concurrently, and a single patient may receive care from several specialists.

Patient has total participation in Admits: a patient record is only created when they are admitted, so every patient must be linked to a ward. Ward has partial participation in Admits: wards are set up in the system before patients arrive, so a ward can exist with zero current patients.